

ADHAM HEWALA

Backend & AI Engineer · ERP & Manufacturing Systems

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PROFILE

Backend engineer specialising in ERP and manufacturing systems - and the AI layers built on top of them. Since 2024: Python and FastAPI services behind RFID-driven inventory, production tracking, purchasing and order management for garment and textile manufacturers; led the Egyptian Tax Authority (eTax) integration across a multi-tenant retail ERP. Creator of Lotus Platform, an AI-powered data-analysis product that won the Cairo AI Hackathon 2025 and placed 6th nationwide in DIGITOPIA 2025. AWS Certified Cloud & AI Practitioner. English C1.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS

Backend & APIs: FastAPI, Django, Django REST Framework, REST API design, microservices, asynchronous background processing

ERP & Manufacturing: Odoo (custom modules, migrations, integrations), RFID data capture, inventory and production tracking, POS, eTax integration

AI & Machine Learning: LLM integration, RAG, vector databases, prompt engineering, self-hosted NLP models, scikit-learn, Pandas, NumPy, demand forecasting, reinforcement learning

Data: PostgreSQL, MySQL, Redis, query optimisation, indexing, data migration

Cloud & DevOps: AWS (Certified Cloud Practitioner), Docker, containerised deployment, Linux, Git/GitHub, CI/CD, Postman

Practices: Agile, code review, testing, system design, AI-assisted development (Claude Code, Cursor)

PROFESSIONAL EXPERIENCE

Kulsys Technologies — Software Engineer

Jan 2026 – Present

Cairo, Egypt · On-site

Backend engineer on Scalix (scalix.com), an RFID-powered ERP platform for the garment and textile manufacturing industry.

- Design and ship the REST APIs behind inventory, purchasing, production tracking and order management, serving 15+ manufacturing clients across 30+ production sites.
- Built the RFID ingestion path that turns 200k+ tag reads per day, turning raw RFID events into live stock and work-in-progress visibility on the factory floor.
- Reduced inventory reporting latency from 2.8s to 650ms through query optimisation, indexing and caching.
- Moved releases onto containerised deployment, cutting release time from 30 minutes to under 10 minutes and making rollbacks routine.
- Work directly with production and warehouse teams, translating floor-level constraints into product decisions.

myGINNE — Software Engineer

Oct 2024 – Dec 2025

Cairo, Egypt · Hybrid

ERP and retail management platform built on Odoo and FastAPI, serving 50+ retail clients across sales, inventory and point of sale.

- Led the Egyptian Tax Authority (eTax) integration, automating tax compliance and electronic receipt management for every transaction on the platform, eliminating manual receipt processing for 50+ retailers.
- Designed administrative dashboards and role-based access control used by 100+ users across client organisations.
- Integrated third-party platforms including Zoho to streamline retailer onboarding, subscription management & sales ops.
- Designed and optimised the platform's REST APIs across the core sales and inventory modules.
- Contributed to migrating POS web services to a React Native mobile app, helping extend the platform's retail workflows.

Forge Solutions — Odoo Developer (Part-time)

Jan 2024 – Apr 2025

Dubai, United Arab Emirates · Remote

- Developed and customised Odoo modules for CRM, inventory, sales and operational workflows across 10+ client projects.
- Executed data migrations from legacy systems into Odoo, reconciled against source records.
- Integrated third-party services and automated business processes within client ERP deployments.

Digital Egypt Pioneers Initiative (DEPI), MCIT — Data Science & AI Trainee

Nov 2024 – Mar 2025

Cairo, Egypt · Hybrid · National programme completed part-time alongside full-time employment

- Built and evaluated machine learning models on real datasets using Python, Pandas, NumPy and scikit-learn.
- Graduation project: an AI-based diabetes prediction and clinical decision-support system.
- Awarded the DEPI Certificate of Recognition for top performance in the cohort.

Pinatro — Back-End Developer Intern

Jun 2023 – Aug 2023

Cairo, Egypt · Remote

- Built and maintained backend features and REST APIs for web applications within an Agile, Git-based team.

CCIC — Back-End Developer Intern (Python / Django)

Aug 2022 – Oct 2022

Mansoura, Egypt · Hybrid

- Built a full-stack e-commerce application in Django - product management, user accounts, cart and order processing - with REST APIs in Django REST Framework.

SELECTED PROJECTS

Greeny — Grocery E-Commerce Platform | Django, PostgreSQL, REST APIs

- Full-featured grocery e-commerce platform with product catalog, categories and brands, user accounts, cart, checkout, secure payments and order tracking.
- Personalized shopping experience with AI-powered product recommendations to improve product discovery and help customers find relevant groceries based on their preferences and shopping behaviour.
- Customer-focused features including product comparison, profile management, order history, and a content/blog section supporting the complete online shopping journey.

Inventory Management & Demand Forecasting System | FastAPI, PostgreSQL, Redis, Docker

- Inventory and stock tracking with supplier management, purchase orders, warehouse operations and reporting.
- Demand forecasting on historical sales data to flag potential stock shortages and generate restocking recommendations.
- Improved throughput through database optimisation, Redis caching, asynchronous processing and a containerised architecture.

Job Portal & Recruitment Management System | FastAPI, PostgreSQL, Redis, Docker

- Full-stack recruitment platform: job posting, applications, résumé upload, interview scheduling and application tracking.
- Automated candidate-to-job matching using AI-powered semantic analysis, with role-based access control and advanced search and filtering.
- Tuned backend performance via PostgreSQL indexing, query optimisation, Redis caching and asy background processing.

AWARDS & ACHIEVEMENTS

- Winner — Cairo AI Hackathon 2025. Lotus Platform, EGP 100,000 prize.
- 6th Nationwide — DIGITOPIA 2025, AI & Software Solutions Track, reaching the final stage with Lotus Platform.
- Bronze Medal — Huawei ICT Competition 2025/2026, Cloud Track, National Phase.
- Finalist — AWS DeepRacer Egypt; finished within 1.138 seconds of first place with reinforcement-learning models.

CERTIFICATIONS

- AWS Certified Cloud Practitioner — Amazon Web Services
- AWS Certified AI Practitioner — Amazon Web Services
- HCIA — Cloud Service / AI — Huawei
- AWS AI & Machine Learning Program — Udacity
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI / Coursera
- Advanced Learning Algorithms — DeepLearning.AI / Coursera
- AI Career Essentials (AICE) — ALX
- McKinsey Forward Program — McKinsey & Company

EDUCATION & LANGUAGES

Mansoura University — Faculty of Computers & Information Sciences

2019 – 2023

Mansoura, Egypt

Bachelor of Science in Computer Science.

Languages: Arabic — Native · English — Full professional proficiency (C1) · German — Intermediate (B1)